

Research article

Managing tourism pressure: Exploring tourist traffic patterns and seasonality in mountain national parks to alleviate overtourism effects

Mateusz Rogowski^{a,*}, Bernadetta Zawilińska^b, Joanna Hibner^c

^a Department of Tourism and Recreation, Adam Mickiewicz University, Poznań, Krygowskiego 10, 61-680, Poznań, Poland

^b Department of Social and Economic Geography, Krakow University of Economics, Rakowicka 27, 31 510, Kraków, Poland

^c Institute of Geography and Spatial Management, Jagiellonian University in Krakow, Gronostajowa 7, 30-387, Kraków, Poland

ARTICLE INFO

Keywords:

Tourist traffic
Visitor flow
Overtourism
National park
Tourist seasons
Visitor index

ABSTRACT

Overtourism results in pressure exerted on the natural environment, negative impacts on local communities and lower tourist satisfaction in many protected areas (PAs). This phenomenon has become an increasingly important issue in the management of national parks (NPs) in Poland, especially the mountain ones. Countering and alleviating the effects of overtourism requires developing appropriate management strategies based on the detailed analysis of tourism monitoring data. Having recognized this need, a research was conducted to develop methodology for analyzing daily visitors in NPs (methodological objective), to identify regularities characteristic for visitor flow in mountain NPs in Poland (cognitive objective), and to present the usefulness of monitoring studies in developing effective strategies aimed at tourist traffic management in NPs and their surrounding areas (application objective). The availability of data allowed conducting analyses for three national parks: The Pienniny NP (PNP), The Stolowe Mountains NP (SMNP) and The Tatra NP (TNP).

The analyses covering the period 2017–2022 revealed a strong intensification of visitor flow and a growing problem of overtourism in the analyzed areas, primarily in TNP. An increase in maximum daily values, the number of days showing these values, and intensified tourist traffic off-high season were found. Using Daily Visitor Index and Seasonality Index, high-, mid-, and low seasons were identified, including the similarities in traffic seasonality between PNP and SMNP, along with the different nature of TNP, where the winter season was additionally separated. Using the Monthly Visitor Index and Gini coefficient, the long-term variation analysis revealed a reduction in the seasonality of tourism resulting from higher intensity of visits off summer season, in particular the extension of mid-seasons. Due to the growing number of high season visitors and the increasing visitor flow, it is necessary to implement time-limit measures at specific locations and activities aimed at temporal and spatial tourist traffic dispersion. The study shows the development process of overtourism in NPs. It also gives indications that the uneven spatial distribution of visitor flow determines the need for diagnosing overtourism in places and periods with the highest visitation intensity. Such a diagnosis is essential for effective tourism management in NPs and supports decision-making processes for the development of surrounding localities.

1. Introduction

Modern tourism offers numerous economic and cultural benefits, contributing to regional development and the promotion of natural heritage (Driml and Common, 1995; Buckley, 2010; Worboys et al., 2015; Leung et al., 2018). However, along with the growing popularity of tourism destinations, certain challenges emerge regarding the sustainability of these areas (Mika, 2024). Overtourism has been recognized as one of the main problems currently affecting tourist areas

worldwide (Capocchi et al., 2019; Perkumienė and Pranskūnienė, 2019; Butler and Dodds, 2022; Buitrago-Esquinas et al., 2023). This term describes a situation when an excessive number of tourists exceeds the reception capacity of an area, resulting in negative effects on the environment, local communities and the quality of tourism experience (Zareba, 2010; Capocchi et al., 2019). Many cities, including Barcelona, Dubrovnik, Venice (Seraphin et al., 2018; Hernández et al., 2019; Zemla, 2024) and other destinations, for example, many areas in the Mediterranean basin (Papadopoulou et al., 2023) are struggling with

* Corresponding author.

E-mail address: mateusz.rogowski@amu.edu.pl (M. Rogowski).

<https://doi.org/10.1016/j.jenvman.2024.123430>

Received 2 April 2024; Received in revised form 31 October 2024; Accepted 19 November 2024

Available online 29 November 2024

0301-4797/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

overtourism. Overtourism in national parks has also been noted e.g. in: Saxon Switzerland NP in Germany (Schamel and Job, 2013), Low Tatra NP in Slovakia (Svajda et al., 2018), Norway's fjords (Oklevik et al., 2019), Iceland (Sæþórsdóttir et al., 2020a, 2020b), Plitvice Lakes NP in Croatia (Mandic, 2023) and Stołowe Mts. National Park (Rogowski 2024). However, the evolving contemporary trends related to active recreation in close contact with nature exacerbated by the Covid-19 pandemic (Lebrun et al., 2021; Cheng and Liu, 2022; Rogowski, 2022a, 2022b; Volgger et al., 2022), caused a significant increase in tourism pressure on PAs escalating conflicts between different management objectives. The problem is not limited to the growing number of visitors, but also relates to an uneven temporal distribution of visitor flow, resulting in intensified negative phenomena at certain times (Smolčić Jurdana and Zmijanović, 2014; Walden-Schreiner et al., 2018), which implies the need to develop monitoring, conduct in-depth studies of tourism and its natural and socio-economic impacts in the affected PAs, as well as develop strategies to alleviate conflicts and counteract negative phenomena (Eagles et al., 2002; Ghelichipour and Muhar, 2008; Kohl and Pekny, 2011; Hibner et al., 2023b).

In Poland, the issue of overtourism refers to many PAs, in particular the majority of national parks (NPs) in the Carpathians and the Sudetes (Adach et al., 2023). Unfortunately, conducting constant monitoring of tourist traffic is not common in Polish NPs, moreover, each park has developed its own visitor flow monitoring system, based on different data sources and data acquisition techniques (Zawilińska, 2021). The detailed data on visitor flow broken down by days throughout the year are available only for few NPs; Rogowski, 2020; Prędkie and Demko, 2021; Fidelus-Orzechowska et al., 2024). At the same time, each of these parks conducts measurements and analyses based on the individually adopted methodologies, therefore their results may not be comparable. The lack of knowledge about the temporal visitor flow distribution in Polish NPs, and the respective comparative analyses inspired conducting the presented study, which – due to tourist traffic specificity and data availability – was narrowed down to mountain parks.

The purpose of the study is to develop a methodology for analyzing daily visitors in mountain NPs (methodological objective), to identify tourist traffic regularities (cognitive objective), and to show the usefulness of monitoring studies in developing effective strategies for visitor flow management in NPs (application objective). The research was carried out based on the example of three parks: The Tatra NP (TNP), The Pieniny NP (PNP) and The Stołowe Mountains NP (SMNP).

2. Tourism carrying capacity in protected areas and the problem of visitor flow adequate monitoring

PAs around the world serve an important social function by providing space for recreational purposes. In particular, easily accessible places or top attractions within particular PAs attract tourists generating overtourism (Eagles et al., 2002; Šturcel and Švajda, 2005; Plummer and Fennell, 2009; Buckley, 2010). The problem of an increasing number of visitors in the environmentally valuable areas emerged in the last century and at that time, mainly out of concern for the state of the natural environment, carrying capacity indicators, understood as the maximum number of people who can stay in a given place without damaging the natural environment, started to be established (Cole, 2004; Staniewska-Zątek, 2007; Zaręba, 2010; Hibner et al., 2023a). The theory of carrying capacity is most likely rooted in pastoral ecology and at that time referred to the number of sheep that could feed from a designated piece of pasture (Hausser et al., 2006). The first developed indicators referred exclusively to the environmental impact aspect, however, by the 1960s, J.A. Wagar began to emphasize the social aspect. Carrying capacity according to J.A. Wagar, is not only a characteristic of a place, but depends on the needs and perceptions of visitors and should be adapted to the management plans of the particular area. Therefore, in later studies, carrying capacity indicators were determined not only on the basis of environmental studies, but also based on the research

addressing the visitors' crowding perception as well (Stankey and McCool, 1984; Cole, 2004; Sterl et al., 2004; Hausser et al., 2006; Somarriba-Chang and Wallentinus, 2012). Since the 1980s, the World Tourism Organization has defined tourist carrying capacity as "the maximum number of people that can visit a tourist destination simultaneously, without causing damage to the physical, economic, social, and cultural environment or an unacceptable decline in the quality of tourists' satisfaction with their stay" (Sobhani et al., 2022). EUROPARC Federation distinguishes 3 types of tourist carrying capacity indicators: (1) ecological carrying capacity - which refers to the negative impact on the natural environment; (2) socio-cultural carrying capacity - which refers to the negative impact on the local community and (3) psychological (social) carrying capacity - which refers to the negative impact on tourists and the recreational, aesthetic values sought by them (Zaręba, 2010).

In some countries (e.g. Portugal), the concept of carrying capacity is a formal part of protected area management strategy, but despite that, not every area has developed such an index (Monteiro and Pereira da Silva, 2012). In Poland, since the 1970s, the issue of determining the tourism carrying capacity of natural areas, including national parks, has been under discussion. Numerous scientific studies have been conducted on this topic and carrying capacity indicators have been established for about half of the NPs. However, all of these indicators were set between the 1970s and 1990s, and therefore do not take contemporary conditions into account (Graża-Zwolińska, 2009). Additionally, indicators for the capacity of tourist trails and the capacity of tourism infrastructure within and around NPs were established. Currently, in the national park protection plans (as mandatory documents), the carrying capacity of areas designated for tourism and tourist trails are specified. However, these indicators are based solely on ecological studies and the physical limits of the area. Indicators for socio-cultural and psychological capacity are not established. SMNP is an example of a national park with multifaceted research in this area, where an attempt was made to assess the tourism carrying capacity of its most popular trails. This assessment considered both biotic and abiotic factors of the surrounding landscape, the technical aspects of the existing tourism infrastructure, and social factors related to visitor satisfaction. Hourly visitor data were used to establish limits verified later through trail measurements, observations of visitors' behavior at various levels of visitor traffic, and a satisfaction survey (Rogowski, 2019).

In all national park, particularly the ones heavily visited by tourists and in their surrounding areas, studies should be conducted on the multifaceted impacts of tourism. This monitoring should include both the effects on the natural environment and landscapes (to inform about the determination of the ecological carrying capacity for tourism) and the effects on people (to help define the psychological and socio-cultural carrying capacity for tourism). It is important to emphasize that mass and excessive tourism leads not only to negative changes in the natural environment (Rączkowska and Kozłowska, 2010; Manning and Anderson, 2012; Santoro et al., 2012; Mitterwallner et al., 2023) and landscape (Jančura et al., 2009), but also has negative impact on local communities (Damjanović, 2021; Zawilińska, 2023). Furthermore, the impacts of tourism contribute to the problem of littering (Pickering et al., 2010; Zhong et al., 2011; Hu et al., 2018), especially with plastic waste and other non-biodegradable materials, the amount of which has been increasing recently (Religa and Addach, 2020). In addition, overtourism leads to lower tourist satisfaction with visiting a particular protected area (Belsoy et al., 2012; Ferreira and Harmse, 2014) and increased safety risks for tourists, especially in high-mountain areas. The conducted monitoring studies should result in actions aimed at preventing and mitigating these negative impacts (Prato, 2001).

The efforts to alleviate negative effects of intensive tourism are based on two opposing strategies. The first consists in gathering tourists in a few less sensitive zones of the protected area, thereby relieving the pressure on more anthropopressure-sensitive sites (Gulić et al., 2014). This strategy follows the concept of creating the so-called "honey pots",

i.e. tourist attractions or forms of tourist development that attract and retain tourists. The first strategy refers to the spatial and temporal concentration of tourists (e.g., during holidays), aiming to make part of the year free from tourist pressure, thus providing an opportunity for wildlife regeneration. The alternative strategy involves temporal and spatial deconcentration of tourist traffic, resulting in levelled, however, less intense tourist pressure. This strategy is more beneficial from the perspective of tourist satisfaction as well as local development (it leads to a reduction in tourist traffic seasonality and levelled distribution of economic benefits from tourism in time and space). Reducing visitor flow seasonality is considered by experts from the European Commission for Sustainable Development as one of the crucial tasks for sustainable tourism development across the entire Europe (Smolčić Jurdana and Zmijanović, 2014).

In practice, while aiming at counteracting the negative natural and social impacts and maximizing the benefits for local communities associated with tourist visits in NPs, the elements of both strategies can be combined. Measures are taken to reduce visitor flow seasonality and distribute it more evenly in the localities surrounding NPs, while introducing incentives and restrictions to concentrate traffic in certain places, at a certain time along with channeling it, as well as temporarily excluding certain areas from visitors' access. This can be achieved by implementing (1) spatial limitations, such as an obligation to use the exclusively marked tourists trails or zoning more sensitive areas; (2) temporal limitations, such as limitations of entry during night time or winter season, which foster wildlife recovery process; (3) qualitative limitations, such as limitations for conducting a particular tourist activity, e.g., skiing, climbing, kayaking etc. and (4) quantitative limitations establishing visitor capacity limits (Eagles et al., 2002; Skawiński, 2010; Kohl and Pekny, 2011). The quantitative limitations are considered the most challenging ones because the indicator specifying the exact number of visitors, who may enter the particular area without causing large destruction, is extremely hard to determine. Moreover, determining the carrying capacity indicator itself does not solve the problem caused by overtourism, as visitors are willing to accept the limits if they understand and experience the consequences of overcrowding (McCool and Lime, 2001; Immoos and Hunziker, 2015). Nowadays, there are several theories and concepts supporting sustainable management strategies, such as (1) LAC – Limits of Acceptable Changes – a theory based on both ecological and social aspects; (2) VERP – Visitors Experience and Resource Protection – based on the balance between conservation, visitor satisfaction and quality of experience and (3) VIM – Visitor Impact Management – i.e. reducing the negative impact of tourist movement (Hammit et al., 2015; Hibner et al., 2023b).

In order to implement successful visitor management plan it is crucial to monitor daily visitors. Monitoring fosters management planning by identifying problem areas, reducing potential clashes between the environment and human activity or identifying conflicts between different user groups (Cessford and Muhar, 2003; Kajala et al., 2007). However, the information on annual visitor flow in PA is obviously insufficient. To support effective management strategy it is essential to conduct research on spatial and temporal distribution of visitors. In 2003, Cessford and Muhar divided monitoring methods into 4 main categories: (1) direct observation including: field observers, i.e. the most traditional visitor monitoring method – manual visitor counting, camera recording – based on using camera traps located in different areas, remote sensing – air photos or other images; (2) on-site counters – automated counters; (3) visit registrations and (4) inferred counts – e.g., visitor surveys. Nowadays, the spectrum of available counting techniques is much wider. They include: crowdsourcing, social media analysis, mobile applications, computer vision approach or data from mobile network operators. Moreover, these tools can support gathering data not only about the number of visitors, but also on their spatial and seasonal distribution, and in many cases, also the forms of practiced tourism (Levin et al., 2017; Tenkanen et al., 2017; Muñoz et al., 2019; Erlebach and Romportl, 2021; Hibner et al., 2023b). From the

perspective of PAs managers, having detailed data on spatial and temporal distribution of visitors is crucial for developing appropriate strategies aimed at managing tourist traffic and making informed decisions along with taking actions regarding park accessibility (Eagles et al., 2002; Cessford and Muhar, 2003; Kajala et al., 2007; Zawilińska, 2021).

In Polish NPs, visitor counting is primarily based on the number of admission tickets sold. It should be noted that tickets are often collected only at specific hiking trails or certain attractions and usually during the tourist season. Although some parks additionally use automatic counters, they are placed only along certain tourist trails. Hence, a considerable number of NPs depend solely on estimated data for overall visitor flow and lack specific information on daily visitors covering the entire park area throughout all months of the year (Spychala and Graja-Zwolińska, 2014; Zawilińska, 2021). Comparing to all NPs in Poland, the tourist traffic monitoring system in mountain parks is relatively well-developed. All these parks provide daily data on tourist traffic throughout the year, however, this data is obtained based on various techniques and, in some cases, remains incomplete.

3. Materials and methods

The data were obtained from the directorates of national parks, based on the criteria: (1) minimum five years, in this case 2017–2022 period, (2) daily resolution, (3) entire park area. Out of the eight national parks located in Poland in mountain areas of the Carpathians and the Sudetes, the data following the above criteria were available for three national parks: The Pieniny National Park (PNP), The Tatra National Park (TNP) and The Stołowe Mountains National Park (SMNP) (Fig. 1). The first two are located in the Carpathians. PNP is the oldest (1932) and one of the smallest national parks in Poland (23.5 km²). It covers the central part of Pieniny range, where the main relief features are Dunajec Gorge surrounded by limestone rock walls and the towering peaks of Mount Trzy Korony and Mount Sokolica listed among the most spectacular vantage points in the Polish Carpathians. TNP (211.6 km²) covers the Tatra Mountains, the highest mountain range in Poland and the entire Carpathians. This park is distinguished by its varied high-mountain relief, with sharp ridges and post-glacial basins. In turn, SMNP (63.4 km²) located in the Sudetes is primarily famous for its rock labyrinths formed by the weathering of sandstone rocks. The volume and intensity of daily visitors in these NPs is much higher than the average values for NPs in Poland. All of these parks face serious problems resulting from overtourism.

Each of the analyzed NPs has developed its own methodology for acquiring daily visitors data:

1. PNP collects data in a hybrid manner: based on admission tickets during the tourist season (April–November) at four entry points (Trzy Korony, Sokolica, Czorsztyn Castle, The Pieniny Road) refined based on the park algorithm, and in the remaining period obtained from pyro-electric sensors as well as fees collected from Dunajec Gorge rafting and entry fees for the environmental exhibition at the national park headquarters. The analyses conducted by the park showed that the hybrid method of data collection provides more comprehensive results. During peak tourist seasons, ticket sales data are more accurate than sensor measurements, which can have greater errors in high-traffic situations. However, when ticket sales are not occurring, automatic sensors serve as the sole data source, delivering very accurate measurements. The data from Dunajec Gorge rafting are presented in a weekly resolution, which had to be modeled to daily data. The high correlation (0.94) of the input data for the Trzy Korony and Dunajec Gorge rafting during the tourist season allows modelling the daily data by assuming a weekly proportion of climbs on Trzy Korony.
2. TNP gathers daily data from ticket sales throughout the year, obtained from seventeen entry points and Kasprowy Wierch cable car tickets (which includes admission fee to the national park).

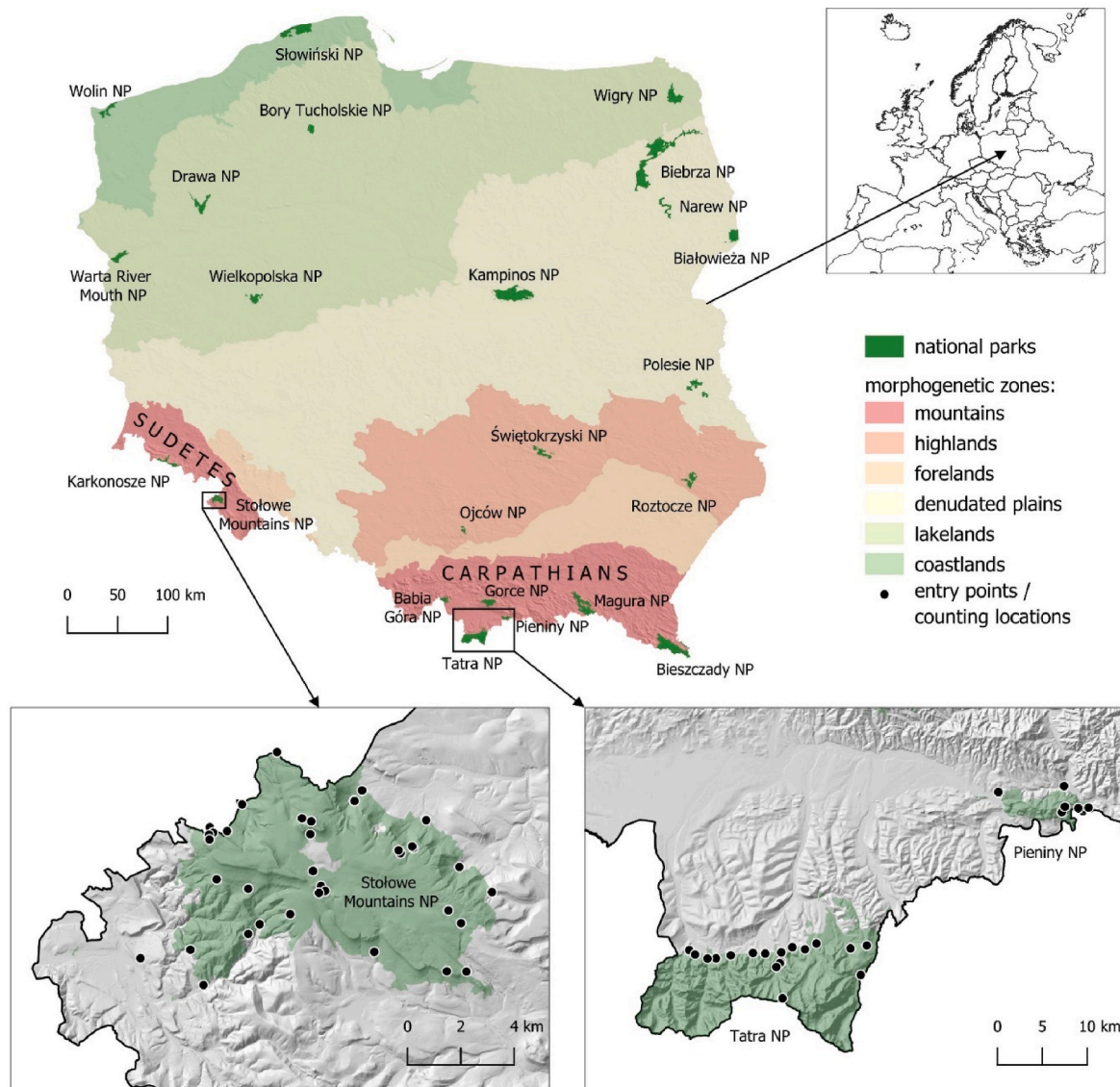


Fig. 1. Location of the analyzed national parks.
Source: own elaboration

Regarding holidays on which ticket offices are closed (January 1, November 1), the authors refined the data based on the average value for the day before and after.

3. SMNP obtains daily data pyro-electric sensors installed at each park entrance functioning as Monitoring System of Tourist Traffic (MSTT) (Rogowski, 2020). Daily data has been calibrated by calibration formula (Rogowski and Piotrowski, 2022) and aggregated. The comprehensive automatic measurement system provides consistent data for the entire park area, which would not be possible based solely on admission fees, which apply only to two trails.

After verifying data quality and their consolidation, a preliminary analysis was performed using the descriptive statistics measures. Next, tourist seasons in each park were determined using Daily Visitor Index (DVI) and Seasonality Index (SI). This was followed by analyzing the long-term variation in tourist traffic based on Monthly Visitor Index (MVI) and Gini Coefficient. Visitor Index is the ratio of daily visitors to the mean value for a given period (Rogowski 2021). Two index variants were adopted in the study:

- 1) Daily Visitor Index (DVI) is the ratio of daily visitors to the average daily visitors per year:

$$DVI = \frac{\sum v_d}{\bar{v}_y}$$

where:

$\sum v_d$ – daily visitors

$\bar{v}_y$ – average daily visitors per year.

- 2) Monthly Visitor Index (MVI) is the ratio of daily visitors per month to the average daily visitors in the period under study.

$$MVI (2017 - 2022) = \frac{\bar{v}_{dp}}{\bar{v}_{ap}}$$

where:

$\bar{v}_{dp}$ – average daily visitors per month

$\bar{v}_{ap}$ – average daily visitors in 2017–2022.

Seasonality Index (SI) is a measure estimating the relationship between the DVI mean of days free of work and the DVI mean for weekdays. It can be interpreted as the ratio of tourist traffic volume during these two types of days. When SI equals 1.00 in the examined season, the number of weekday visitors is comparable to the number of visitors on days free of work. SI greater than 1.00 confirms that weekend tourism is more popular than weekday tourism. SI identifies differences in the average number of visitors on weekends and holidays occurring cyclically in successive weeks. SI describes the strength of the increase in visitors during successive weekends relative to weekdays. SI increase typically occurs in spring and autumn during public and religious holidays. SI formula:

$$SI = \frac{\bar{X} \text{ DVI}_{\text{weekend}}}{\bar{X} \text{ DVI}_{\text{weekdays}}}$$

where:

$\bar{X} \text{ DVI}_{\text{weekend}}$ – DVI mean of days free of work

$\bar{X} \text{ DVI}_{\text{weekdays}}$ – DVI mean of weekdays

Gini coefficient is used to measure the inequality of goods distribution, primarily household income distribution, and therefore it is often referred to as the social inequality index. However, due to its structure, based on the Lorenz curve, it can also measure the inequality of visitor seasonality. Gini coefficient takes values in the range (0; 1). Gini coefficient is less sensitive to peaks but more sensitive to off-peak changes (Þórhallsdóttir and Ólafsson, 2017). The presented formula takes the following form:

$$G = \frac{2}{n} \sum_{i=0}^n (Xi - yi)$$

where:

n = number days.

X_i = rank of days 1/365, 2/365 ... etc. So $x_i = i/n$

y_i = cumulated fractiles in the Lorenz curve.

DVI and SI identify tourist seasons. DVI for a day of the year classifies it into one of three types of seasons: high, mid- and low season. The division is made using thresholds: (1) high season $DVI > 1.5$; (2) mid-season $1.5 \geq DVI \geq 0.5$; (3) low season $DVI < 0.5$. During each type of season, short periods and single days showing deviations from the above distribution are possible. $SI > 2.0$ informs about at least twice as many visitors on weekdays of a given week than at weekends, which qualifies the period as mid-season. $SI < 2.0$ informs about at least twice as many visitors during weekends than on weekdays of a given week, which qualifies the period as high season and low season.

4. Results

4.1. General regularities in tourist traffic

The analyzed NPs are characterized by very high annual visitor flow (Table 1), significantly exceeding the average for Polish NPs (696 000 visitors in 2022). For TNP, it reached 4.5 million people in 2022, making it one of the most visited NPs in Europe. Regarding the analyzed parks area, such high attendance results in severe tourist overload: 463 tourists per ha in PNP, 226 tourists in TNP, and 205 tourists in SMNP, while the average for Polish NPs was 50 tourists per ha (Environment, 2023).

The analyzed NPs are characterized by an uneven distribution of daily visitors throughout the year and days with very high and very low turnout. It is confirmed by large differences between the highly inflated mean and median. They result from very high maximum values on several days representing annual climaxes. Uneven annual distributions of daily visitors are confirmed by positive values of skewness and kurtosis making them right-skewed and leptokurtic with more high extreme values than in a normal distribution. This element, along with the median increase (3-fold in TNP, 2.5-fold in SMNP and 1.7-fold in PNP) mean that the number of days recording very high numbers of visitors increases each year. This exacerbates the problem of overtourism. Uneven annual distribution of daily visitors is further confirmed by high coefficient of variation and standard deviation, providing an

Table 1
Statistics of daily visitors in national parks.

Measure	2017	2018	2019	2020	2021	2022
PNP						
Annual total	924,677	992,712	973,654	821,306	1,077,443	1,095,074
Mean	2533	2720	2668	2244	2952	3000
Median	855	1377	1246	626	1098	1432
Maximum	15,096	12,852	14,490	12,964	14,900	12,867
Kurtosis	1.7	0.3	1.1	1.2	0.8	0.4
Skewness	1.5	1.1	1.4	1.6	1.4	1.2
Standard deviation	3308	2998	3134	3121	3670	3413
Coefficient of variation	131%	110%	117%	139%	124%	114%
TNP						
Annual total	2,291,054	2,789,356	2,842,939	2,140,680	3,153,617	4,518,015
Mean	6277	7642	7789	5849	8640	12,380
Median	3052	4859	4598	3501	4802	9000
Maximum	39,512	36,060	44,976	27,769	44,301	47,403
Kurtosis	2.4	1.0	2.0	0.8	1.4	1.2
Skewness	1.7	1.3	1.5	1.3	1.5	1.3
Standard deviation	7835	7025	7755	6437	9350	9901
Coefficient of variation	125%	92%	100%	110%	108%	80%
SMNP						
Annual total	872,959	1,065,279	913,736	917,525	1,276,147	1,304,654
Mean	2386	2913	2498	2507	3491	3569
Median	918	1508	1478	846	1854	2255
Maximum	14,822	15,946	17,229	14,654	20,998	16,720
Kurtosis	2.0	1.3	2.9	1.4	1.5	0.7
Skewness	1.5	1.3	1.5	1.5	1.5	1.2
Standard deviation	2803	3270	2713	3269	4088	3782
Coefficient of variation	117%	112%	109%	130%	117%	106%

Source: own elaboration based on the data from national park directorates.

opportunity to delimit tourist seasons of the year. It allows determining the duration of high season and the highest daily values accurately. The year 2020 was atypical due to the Covid-19 pandemic restrictions. The most dynamic increase in visitor flow occurred at the beginning of the high season, coinciding with the gradual lifting of restrictions in the spring. A significant intensification of visits was observed during the summer. Additionally, a higher visitor flow was noted in the mid-season, particularly in the autumn. As a result, the most popular national parks experienced varying impacts: TNP saw a large decrease in visitors, as confirmed by Fidelus-Orzechowska et al. (2024), PNP had a slight decrease, while SMNP recorded a small increase. This trend was driven by the need to avoid the more crowded national parks, making visitors opt for less popular ones (Rogowski and Piotrowski, 2022). As a result, after lifting the Covid-19 restrictions, more national parks are affected by overtourism, both the most popular and the less popular before the Covid-19 pandemic. TNP presents the highest visitor growth rate, which along with an increasingly lower coefficient of variation causes high levels of tourism last the longest. Therefore, quiet periods defined as low season disappear. SMNP and PNP are characterized by a similar nature of visitor flows, also confirmed by the measures in Table 1.

4.2. Seasonality of tourist traffic

4.2.1. Spring mid-season

Spring mid-season in PNP and SMNP begins in March, when $SI > 2.0$. During this time, an increase in weekend activity (Friday–Sunday) is recorded, when the number of daily visitors on weekends is 2–4 times higher than on weekdays. Spring mid-season ends with the Long Labor Day holiday weekend when the largest increase in annual visitor flow and sustained high values thereafter are recorded. Spring Mid-season in TNP begins at the end of April, due to the altitude of the Tatra Mts. and longer winter conditions. During the Long Labor Day and the Long Corpus Christi holiday weekends, significantly higher daily visitors are recorded, but after they finish, daily visitor values decrease. More visitors are observed during the consecutive weekends in May and June than weekdays, causing $SI > 3.0$. Therefore this period is classified as mid-season. This situation continues until the commencement of summer holidays in the second half of June, ending the spring season.

4.2.2. High season

High seasons in PNP and SMNP are recorded at the start of the Long Labor Day holiday weekend. After that, the number of visitors remains at a similar level as the school tour season begins on weekdays, and the high number of individual visitors during the following weekends continues. High DVI is recorded for the Long Corpus Christi holiday weekend. SI gradually declines to 0.8–1.4 in July and August. DVIs reach their highest values (>3.00), with a frequent peak during the Long holiday weekend of the Assumption of the Virgin Mary (August 15). After that, visitor flow gradually drops entering the declining phase. High seasons end at the beginning of September, which coincides with the commencing school year. During this time, DVI decreases to 1.5 and SI rises above 2.0 marking a clear difference between daily visitors on weekends and weekdays. High season begins in TNP in late June and early July. At that time $DVI > 1.5$ and $SI < 2.0$, record the highest annual visitor flow with the climax at the Long holiday weekend of the Assumption of the Virgin Mary. After that, visitor flow gradually drops, but high values are also recorded in September. The end of high season begins in late September. DVI falls to approx. 1.0 and $SI > 3.0$. In 2022 a unique situation occurred in TNP, as between August 17 and 25 a drop in DVI from 3.8 to 1.1 was observed, with a simultaneous rise in SI above 2.0, which means a shortened high season due to bad weather, with the first snowfall recorded already in September 2022. However, this situation should be considered unusual.

4.2.3. Autumn mid-season

At the beginning of September, SI shows a significant increase,

exceeding 6.0, in PNP and SMNP. During this time, DVI is highly variable reaching values lower than 1.0, and on some weekend days exceeding 2.0. However, a progressive decline in DVI during subsequent weekends is observed, until late October and early November, when values higher than 1.5 are recorded for the last time in the year. After that visitor flow shrinks ending mid-season in autumn. In PNP and SMNP, mid-season in the autumn of 2017 lasted until late October/early November, to be extended in subsequent years until the first decade of November. Starting in 2021, mid-season in autumn continued till the end of the Long Independence Day holiday weekend (November 11). In 2022 DVI reached the high season value (2.9). Mid-season in TNP begins in late September, when DVI and SI reach similar values as those recorded in early September in PNP and SMNP. In TNP, mid-season in autumn lasts until the Christmas – New Year period. The highest DVIs are recorded during October weekends, only to drop to approx. 0.6–1.0 during November and December weekends. SI reaches high values even in December, thus it cannot be classified as low season.

4.2.4. Low season

In subsequent years, the start of low season in PNP and SMNP is shifted. Daily visitors on weekdays fall in the first quartile, and at weekends in the first or second quartile. Although $SI > 2.0$, it is the low or very low DVI which determines this period qualified as low season. Higher visitor flow in both parks is observed during the Christmas – New Year period, while in SMNP during the periodically organized Winter Half-Marathon on the third weekend of January. DVI and/or SI exceed the figures defined for this period. For both areas, it is the quietest period with a clear decrease in daily visitors.

4.2.5. Winter season, early spring season (TNP)

In TNP, the first four months of the year are characterized by different DVI and SI values comparing to SMNP and PNP, thus identifying a separate period. Winter season in TNP begins in the Christmas – New Year period with an increase in DVI. High DVI values, along with low SI ones are recorded till the end of winter holidays.

Early spring season is a continuation of winter season, characterized by high SI seasonality (3.0–8.0), the lowest daily visitors per year, and the occurrence of winter conditions during early spring, lasting until the end of April, when a large increase in visitors during Long Labour holiday weekend is noticed. In subsequent years, an increase in days with $DVI > 1.0$ is observed, in 2021 three times as many as in 2017 were recorded. This shows the trend of winter season lengthening till March and the shortening of early spring season. It results from the increasingly better conditions for practicing winter tourism in TNP in March and April.

4.3. Changes in daily visitors

In the next step, MVI was used to show monthly variation of the analyzed phenomenon. The highest MVI (2017–2022) was recorded in August for all parks, which was the culmination of high season, while the lowest during the autumn-winter period in PNP and SMNP (December, January) and in March in TNP. In the pre-pandemic period, higher rates were recorded for PNP and SMNP in spring, especially in May, due to the culmination of school trips. In 2022, MVI values for the spring months partially recovered from pandemic losses. Since 2020, MVI maximum values were increasing, and in 2022 they exceeded 3.0 for the first time, which is 3 times the average daily visitors in 2017–2022. These record values indicated the growing problem of overtourism in all national parks (See Figs. 2 and 3).

In TNP, MVI was significantly higher in all months of 2022 comparing to the beginning of the studied period (Table 2), which confirms the growing problem of overtourism. In the pre-pandemic period, $MVI > 1.0$ was recorded for 4–5 months, indicating above-average values in that period. The pandemic year was characterized by lower MVI, including the maximum in August, reflecting attempts to

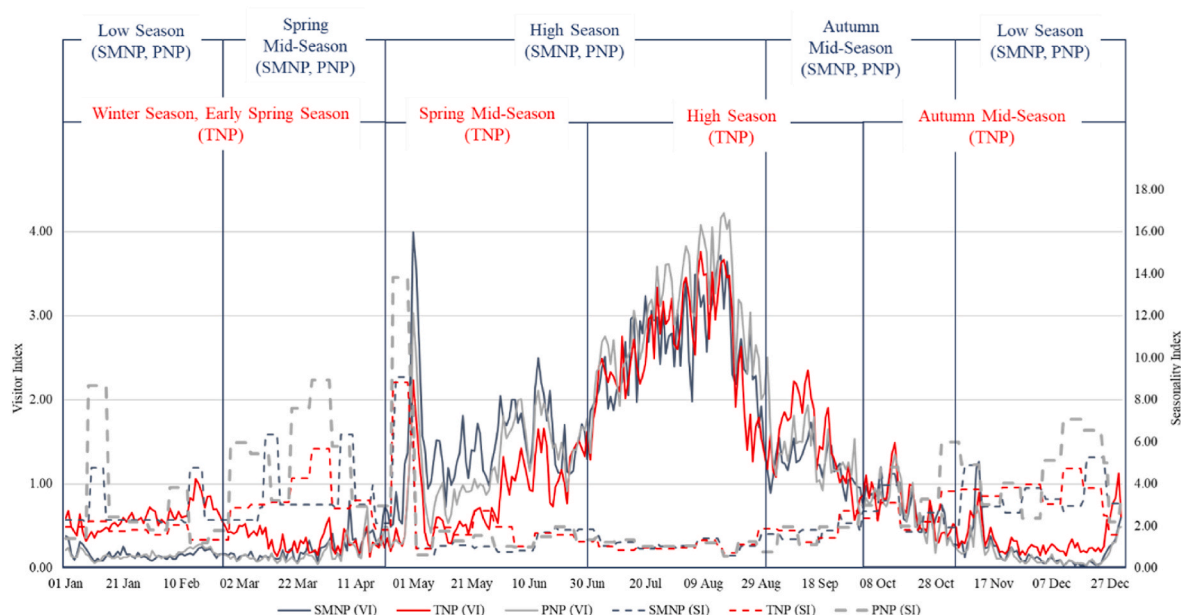


Fig. 2. Seasonality of visitors flow in Stołowe Mts. NP, Pieniny NP and Tatra NP.

Source: own elaboration

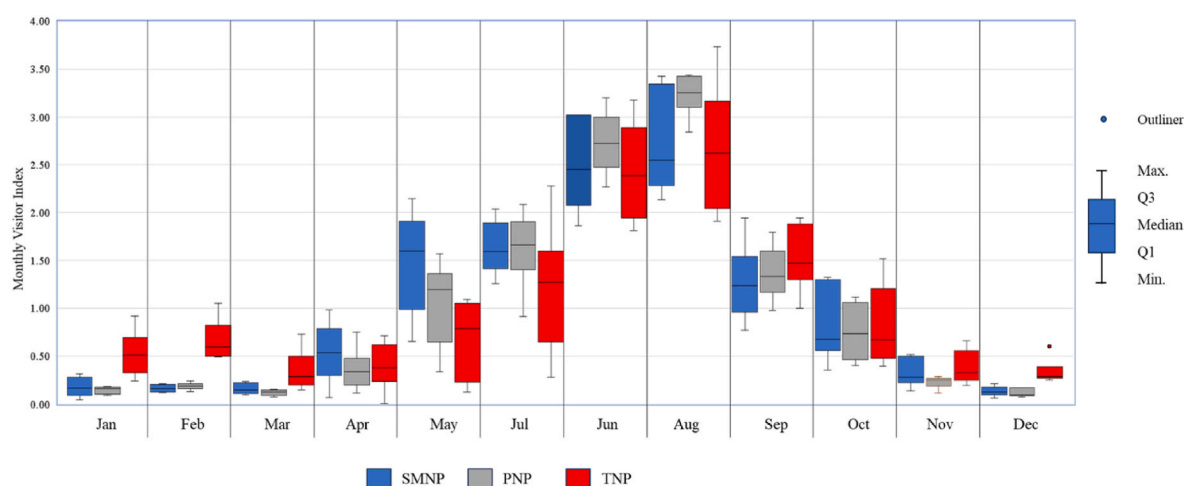


Fig. 3. Monthly Visitor Index of 2017–2023 period in Stołowe Mts. NP, Pieniny NP and Tatra NP.

Source: own elaboration based on the data from national park directorates.

Table 2

Monthly visitor index (2017–2022) for the TNP.

Month	2017	2018	2019	2020	2021	2022
January	0.36	0.45	0.24	0.56	0.62	0.92
February	0.49	0.50	0.57	0.62	0.74	1.05
March	0.22	0.26	0.32	0.15	0.42	0.73
April	0.35	0.59	0.40	0.01	0.31	0.71
May	0.13	1.09	0.87	0.26	0.71	1.04
June	1.37	1.19	1.35	0.77	0.28	2.28
July	2.48	1.99	2.30	1.81	2.79	3.18
August	1.91	2.48	2.76	2.09	2.97	3.74
September	1.00	1.49	1.40	1.45	1.95	1.86
October	0.50	0.62	0.73	0.40	1.10	1.52
November	0.19	0.34	0.26	0.31	0.52	0.66
December	0.28	0.27	0.30	0.25	0.32	0.60

Source: own elaboration based on the data from national park directorates.

avoid infection resulting from the surrounding crowds of tourists. Since 2021, a higher dynamics of MVI increase was observed, including the record high maximum in 2022. Visitor flow in August 2022 was almost 4 times higher than the average value for 2017–2022. In 2022, the period with above-average values was extended to seven months and, for the first time, MVI > 1.0 was recorded for February, whereas the values for January were close to 1.0. This shows the development trend of the winter high season. Having analyzed the minimum MVI, it should be noted that the previously observed most quiet period disappeared, as in 2022 the minimum MVI values for December and November reached 0.6, recorded in mid-season.

PNP experienced an increase in MVI (2017–2022), although lower than in TNP, but the MVI maximum remained above 3.0 (Table 3). This shows that the tourism climax dominated in August, making over-tourism the most acute problem. The period with above-average values was extended in 2017–2022 from four to six months. In the pandemic year, declines were recorded only in spring. In addition, MVI values for September increased in 2020 and autumn 2021, which was evaluated as an

Table 3
Monthly visitor index (2017–2022) for PNP.

Month	2017	2018	2019	2020	2021	2022
January	0.09	0.18	0.10	0.16	0.18	0.17
February	0.13	0.20	0.19	0.17	0.18	0.24
March	0.09	0.13	0.11	0.07	0.15	0.15
April	0.38	0.75	0.35	0.11	0.23	0.33
May	1.27	1.57	1.11	0.34	0.75	1.29
June	1.64	1.57	1.68	0.91	1.84	2.08
July	2.75	2.27	2.54	2.69	3.20	2.93
August	3.18	3.02	3.25	3.25	3.43	3.42
September	0.98	1.53	1.25	1.42	1.79	1.23
October	0.49	0.69	0.78	0.40	1.04	1.12
November	0.11	0.24	0.29	0.26	0.22	0.26
December	0.09	0.10	0.17	0.17	0.07	0.08

Source: own elaboration based on the data from national park directorates.

alternative to summer. In 2022, the trend of above-average values reached in September and October was consolidated, thus extending the autumn period with above-average values. Despite the aforementioned changes, the minimum MVI values indicate that the quietest period was not shortened. $VI < 0.5$ is recorded every year from November to April, which proves the occurrence of a period characterized by low tourist intensity.

In SMNP, MVI values (2017–2022) increased for each month by approx. 50% (Table 4). In the pre-pandemic period, the maxima in July and August exceeded 2.1. In the pandemic year, similar MVI changes in SMNP and PNP were observed. From 2021 onward, similarly as in PNP, MVI values for September and October were above-average. The period featuring above-average values in 2017–2022 was lengthened from four to six months. The analysis of minimum MVI values shows that the quietest period ($MVI < 0.5$) was shortened. In 2017, it covered the first three and last three months of the year, while in 2022 – also the first three months and December.

Gini coefficient for daily visitors regarding all national parks ranged, in 2017–2022, between 0.43 and 0.64 indicating average seasonality (Table 5). The lowest seasonality (G mean values in 2017–2022 = 0.53) occurred in TNP, resulting from the short and still shortening low season. The differences for the lowest daily visitors in low season and the highest daily visitors in high season were the smallest. In PNP and SMNP, the average seasonality indices were slightly higher (0.57 and 0.59), confirming the occurrence of periods with low levels of visitor flow in both parks. In the analyzed period, a gradual decline in Gini coefficient was observed in all parks. For TNP, the highest values were recorded in 2017 and, except for one episode, in 2018. In subsequent years, Gini coefficient was decreasing to reach its minimum in 2022. The obvious exception was 2020, when the coefficient increased due to sanitary restrictions during the pandemic. Its decline indicates a reduction in the degree of tourism seasonality, i.e. the alignment of daily visitors throughout the year. It results from smaller differences between the minimum and maximum values reached in the consecutive years as a

Table 4
Monthly visitor index (2017–2022) for SMNP.

Month	2017	2018	2019	2020	2021	2022
January	0.12	0.11	0.04	0.21	0.27	0.32
February	0.14	0.13	0.12	0.21	0.20	0.18
March	0.13	0.10	0.17	0.11	0.22	0.23
April	0.44	0.98	0.72	0.07	0.37	0.63
May	1.83	2.14	1.51	0.66	1.10	1.69
June	1.52	1.67	1.46	1.26	1.85	2.04
July	2.15	2.20	1.87	2.70	3.02	3.02
August	2.13	2.46	2.33	2.63	3.42	3.31
September	0.77	1.22	1.02	1.40	1.94	1.24
October	0.36	0.68	0.68	0.62	1.32	1.29
November	0.14	0.25	0.25	0.32	0.50	0.51
December	0.11	0.06	0.11	0.14	0.17	0.21

Source: own elaboration based on the data from national park directorates.

Table 5
Gini coefficient (2017–2022) for daily visitors in national parks.

National park	2017	2018	2019	2020	2021	2022
PNP	0.61	0.53	0.55	0.61	0.57	0.53
TNP	0.61	0.49	0.52	0.57	0.55	0.43
SMNP	0.60	0.59	0.57	0.64	0.60	0.56

Source: own elaboration based on the data from national park directorates.

consequence of higher visitor flow growth rate in off-seasons compared to high season.

5. Discussion

Overtourism in certain parts of Poland's mountain NPs represents one of the greatest threats to nature conservation and a major challenge to park management. In response to this situation, an interest in effective tourism management and measures to alleviate its negative effects become vital. The necessity to counteract the negative impacts of overtourism is widely discussed around the world (Chaney and Seraphin, 2023). The problem is even more important considering the significant increase in tourism after the pandemic. It is worth noting, however, that an increase in tourists visiting NPs in Poland was also observed earlier, and only recently its dynamics has intensified (Krakowiak, 2000; Partyka, 2010; Miazek, 2019; Environment, 2023). Similar trends were true for other countries due to changes in tourism preferences, including the pandemic-related increase in the domestic tourism popularity (Lebrun et al., 2021; Volgger et al., 2022). McGinlay et al. (2020) emphasize that managing increased tourism pressure is one of the key challenges for PAs after the pandemic. Effective management requires systematic monitoring of visitor flow and decision-making based on reliable data. Long-term measurement series with daily resolution are highly useful, as they offer the possibility of tracking trends in tourist traffic, finding detailed correlations and predicting daily and hourly visitors for specific days and periods. Daily visitors data allows for the identification of four seasons in NPs: high season, two mid-seasons and low season. Nevertheless, in the parks offering winter tourism the second high season occurs in winter. These areas are particularly vulnerable to the long-term effects of overtourism. Tourist seasons are shifting due to factors like climate change, pandemic-related restrictions, and evolving tourism trends. Recently, summer overtourism has intensified, requiring further diagnosis. The autumn shoulder season now attracts visitor numbers comparable to the high season, while winter sees the greatest increase, eliminating the quiet period. This is reflected in the declining Gini Index. However, hourly visitors data describe the intensity of visitor flow and allow for the calculation of tourism carrying capacity and defining overtourism. The monitoring system for daily visitors should be based on 3 spatial levels (Hibner et al., 2023b). The local level covers a selected tourist trail or attraction. Such monitoring is often implemented in problem areas, e.g., those with excessive visitor flow. SMNP can serve as an example, where detailed visitor flow monitoring supported by the system of sensors allows identifying overburdened places (Szczeliniec Wielki and Błędne Skąły), which results in efforts aimed at dispersing tourists to less popular areas with capacity reserves to absorb tourism (Rogowski 2024). Higher spatial level in the monitoring system covers the entire NP area. Tested solutions in this regard have been implemented in, e.g.: The Karkonosze NP (Czech Republic, automatic counters, Erlebach and Romportl, 2021), Bavarian Forest NP (Germany, based on VGI data, Horst et al., 2023); Montserrat Natural Park (Spain, automatic counters, Farias-Torbidoni et al., 2023); Natura 2000 Ebro River Delta (Portugal, based on VGI data, Jurado Rota et al., 2019); SMNP (Poland, automatic counters, Rogowski, 2020); Danube Floodplains NP (Austria, mixed methods, Arnberger and Hinterberger, 2003). A good example is the tourism monitoring system in the Czech Republic, covering not only NPs, but also the country major hiking and biking trails. The location of automatic traffic sensors and

measurement data are publicly available on an online platform (Měření návštěvnosti, 2024). Proven solutions in this matter can also be found in NPs in the US (mixed methods; Jenkins et al., 2023) and NPs in New Zealand (mixed methods; Cessford and Burns, 2008).

The literature on tourism monitoring highlights the need to implement advanced methods and maintain continuity of measurements (Hornback and Eagles, 1999; Job et al., 2021). The problems related to inadequate monitoring of visitor numbers and data comparability were highlighted in the conducted study. Out of the eight mountain national parks in Poland, it was possible to obtain comparable data on daily attendance from only three of them. Furthermore, the data comparability was determined solely based on the general criteria outlined in the methodology. However, a detailed analysis of the data collection methods reveals certain discrepancies. Primarily, automatic sensors count everyone entering the park, including employees, residents, and other individuals exempt from fees. TNP includes fee-exempt individuals in its statistics but does not account for those entering for work-related reasons. In turn, PNP includes in its attendance figures the visitors to museum exhibitions, which TPN does not consider. The rapid development of electronic ticket sales observed in Poland indicates that permanent ticket collection points will eventually cease to operate. Preparing for this change, parks should enhance their automatic monitoring systems. In this situation, it is recommended to take coordination measures regarding the methodological solutions being implemented. Unfortunately, advanced methods generate costs too high for some NPs to afford. As a result, in most Poland's NP's visitor flow measurements are based on admission tickets sales, but fees are collected only at certain times and at particular entry points. The detailed data have limited temporal and spatial coverage. Annual visitor flow in such cases is estimated based on the available data. Currently, this situation changes following the obligation introduced by successive NPs to pay admission covering the entire park through an electronic ticketing system. In addition to the lack of detailed data, there are methodological discrepancies and ambiguities, with only few studies discussing the monitoring methodology (Prędko, 2012; Prędko and Demko, 2015; Prędko and Demko, 2018; Rogowski and Małek, 2016; Rogowski, 2020). Problems with data availability and comparability prevent preparing comparative analyses between different NPs (Spychala and Graja-Zwolińska, 2014; Zawilińska, 2021). The diversity of spatial and organizational determinants makes developing a universal method for measuring visitor flow in NPs impossible, however, it is advisable to prepare general recommendations and a methodological framework allowing to maintain certain standards and conduct comparative analyses covering many parks (Zawilińska, 2021). Examples of such guidelines can be found in other parts of Europe, including Nordic and Baltic countries (Kajala et al., 2007) and also Germany (Job et al., 2021).

The availability of visitor flow data is an essential tool to support the decision-making process of NP managers. Understanding the variation and seasonality of traffic enables the identification of problem areas and periods, so important for developing more effective tourism management strategies. The studies conducted in three mountain NPs in Poland have shown very high annual visitor flow and a significant intensification of tourist traffic. Concerns are raised by the increase in daily maximums and the number of days with extremely high visitor flow leading to the natural environment and tourist infrastructure overload as well as visitor discomfort. Furthermore, a clear intensification of tourist traffic in off-high season is noticeable. Similar regularities were observed in the Bieszczady PNP (Prędko and Demko, 2021). The periods of the identified tourist seasons occurrence and variation over time were similar in PNP and SMNP, whereas different in TNP. The high-mountain nature of this national park and its winter tourism profile mean the absence of low season typical for other two parks, but the winter season is characterized by an increasing volume of tourist traffic. In turn, low season falling in early spring shows a gradual decline. Similar patterns occur in other mountain NPs particularly attractive for practicing winter tourism, including downhill skiing and skitouring. The occurrence and

length of the season depends on the prevailing winter conditions. For example, as reported by (Erlebach and Romportl, 2021), in the Czech Karkonosze Mountains the most intensive tourist traffic is recorded precisely in the high winter season. The intensification of off-season visits identified in the analyzed NPs, and the gradual disappearance of periods with very low tourist turnout, can be evaluated in two ways. On the one hand, these changes result in the reduction of tourist traffic seasonality, which is beneficial to local economic development by providing opportunities for generating income from tourism throughout the year, stabilizing employment and maintaining continuous services (e.g., transport links, retail and catering offers). However, for the animate nature it means the highest tourist pressure during the previously quiet periods.

Given the very high variation of visitor flow in the analyzed NPs, and primarily the strong concentration of visitors in high season and the increasing number of days with extreme values, an appropriate strategy to counteract overtourism seems reducing visitor flow seasonality and dispersing it during peak periods. "Soft" and "hard" forms of management can serve this purpose (Manning and Anderson, 2012; Hibner et al., 2023b). "Soft" forms include promoting low-season visits, providing information on the current number of tourists staying in a given location, diversifying admission fees and other charges (e.g., parking fees), educational campaigns promoting pro-environmental behavior, and preparing appropriate infrastructure resulting in the desired temporal and spatial alterations (including "honey pots" attracting and retaining tourists). Another solution is applying "hard" forms of management, i.e. implementing temporary entry bans or visitor limits, e.g., reservation systems to enter certain places, or parking lots. In the current situation experienced by some of Poland's mountain NPs and in the face of increasing tourist traffic, using "hard" forms of management seems inevitable. SMNP developed a new approach to the Method of Overtourism Optimization inspired by The Visitor Experience and Resource Protection, The Limits of Acceptable Change and Visitor Impact Management concepts (Rogowski 2024). In-depth studies have shown that visitor flow is unevenly distributed, making it impossible to exceed the tourism carrying capacity of SMNP. Therefore, it is necessary to conduct more detailed hourly visitor measurements at the most overcrowded locations. Exceeding the tourism carrying capacity confirms the presence of overtourism, which is essential for its diagnosis. The diagnosis of overtourism should be carried out in the part of the NP featuring higher concentration of visitors. The occurrence of overtourism was confirmed when the trail's tourism carrying capacity was exceeded. Hourly visitor data is essential to define the tourism carrying capacity of trails.

Mitigation of overtourism is done through Actions to Manage Overtourism. Planning involves stakeholder groups, who select the most optimal and effective proposals from a range of options, adapting them to the specific natural conditions and infrastructure of the NP. In SMNP the entry limits have been introduced in two most popular spots: Szczeliniec Wielki and Błędne Skały, new attractions in and around the park have been made available and alternative sites are being promoted. This has led to a dispersal of visitor flow and a its intensity reduction in the most popular areas (Rogowski, in press). In TNP, paid car parks have been provided with advance booking required and the number of tickets sold electronically corresponds to the number of car park spaces. At PNP, viewing platforms have been installed, thus limiting the number of people remaining the summit at the same time (Trzy Korony viewing platform). Conducting measurements of daily visitors to NPs combined with historical weather data (Semczuk, 2012) provide the opportunity to develop predictive visitor flow models. In addition, if the park provides daily spatial distribution data, models can show the expected visitor flows in spatial and temporal distribution. Access to such forecasts allows better preparation of NPs, adjusting infrastructure and services (e.g., catering facilities) to the expected number of tourists at a given place and time. Modeling results can also be used to disperse visitor flow, e.g., by making public the number of people expected at a

given place at specific times, or at what times particular parking lots are expected to be full. In addition, continuous measurements of tourist traffic should be supplemented by periodic surveys allowing models to be expanded by qualitative issues. In particular, regarding visitor profiles and preferences, as well as the economic impact of tourism (Zawilińska, 2021; Zawilińska and Mika, 2023). Such models will not only help to manage NPs but also facilitate the development of surrounding localities, thus increasing community support for the NP functioning.

6. Conclusions

The research addressing Poland's mountain NPs has shown the lack of complete and comparable daily data on tourist traffic despite the advances in its monitoring. The analyses of measurement data covering three parks showed that different data collecting methodologies are not a limitation in applying the proposed method, which based on long-term measurement series with daily resolution, allows diagnosing time distributions and changes in visitor flow, defining tourist seasons and periods with exceptionally high visitor turnout, crucial in the context of countering overtourism. The aforementioned issues occur in NPs globally. What is missing is a consolidated methodology for monitoring, enabling the sharing of data and experiences related to visitor flow monitoring.

A strong intensification of visitor flow was observed in all three analyzed NPs during the period 2017–2022. A growing problem of overtourism was identified, manifested by an increased number of days with above-average and extremely high visitor flow, an increase in daily maximums, as well as intensified off-peak season tourist traffic. The highest growth rate was recorded in TNP, where visitor flow in 2022 was almost double comparing to 2017. Mass tourism in TNP and its rapid growth rate result in the most noticeable consequences of overtourism. The biggest issue are the long queues of visitors waiting to cross the exposed sections of ridge and slope trails, e.g., Orla Perć. The exposed summits of popular peaks (e.g., Giewont, Świnica, Rysy), where large numbers of visitors gather, have also become very dangerous. SMNP and PNP have the same problems but on a smaller scale. Tourist traffic in all surveyed NPs is characterized by high variation and seasonality, allowing to distinguish high-, mid-, and low seasons. High season occurs in summer, when the highest values falling mainly into the top quartile are recorded, regardless of weekdays, weekends or holidays. Mid-season takes place twice in spring and autumn, when the number of visitors falls into the second or third quartile, with strong variation occurring at the same time. Low season, with small visitor flow (lower quartile) and small variation, falls in the winter months. The exception is TNP, where, due to winter advantages, the winter season can be distinguished, whereas low season, falling in this park in early spring, is gradually disappearing. This situation leads to the development of overtourism, which should be thoroughly diagnosed, and the measures aimed at its mitigation should be agreed upon with stakeholder groups. The identified strategies for mitigating overtourism, including the Method of Overtourism Optimization, can be useful in other regions around the world.

The identified trends in tourist traffic show that further growth in average daily and maximum visitors can be expected, along with the lengthening of mid-seasons at the expense of low season in all analyzed parks. The intensification of visitor flow off summer season and the narrowing gap between high and low season visits are beneficial from the perspective of local economic development, but result in extended time of increased pressure on nature. In turn, intensified high season visits and more extreme daily tourist traffic situations require measures reducing visitor flow to specific destinations during their peak popularity and disperse tourist traffic in spatial and temporal terms.

Further research should focus on predictive modeling based on daily visitors data and weather conditions. Such predictions can support managing tourist traffic, allowing the development of appropriate

operational strategies, improved park services and infrastructure for the number of tourists expected at a particular place and time, thus facilitating the decongestion of visitor flow. This, in turn, will help reduce the negative impact of tourism on the natural environment, while ensuring a positive experience and comfort for visitors. Predictive models of visitor flow can be combined with survey data to enhance the management of NPs and support decision-making processes influencing the development of surrounding localities. Such surveys can not only enable the mitigation of overtourism, but also contribute towards public support for the functioning of NPs.

CRedit authorship contribution statement

Mateusz Rogowski: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bernadetta Zawilińska:** Writing – original draft, Investigation, Conceptualization. **Joanna Hibner:** Writing – original draft, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This article was funded by Adam Mickiewicz University, Initiative of Excellence - Research University (038/04/NP/0005) and Krakow University of Economics (038/GGG/2023/POT). We would also like to express our sincere gratitude to the Pieniny, Stołowe Mountains and Tatra National Parks for providing the data necessary for the preparation of this article.

Data availability

The data that has been used is confidential.

References

- Adach, S., Wojtkowska, M., Religa, P., 2023. Consequences of the accessibility of the mountain national parks in Poland. *Environ. Sci. Pollut. Res.* 30, 27483–27500. <https://doi.org/10.1007/s11356-022-24197-w>.
- Arnberger, A., Hinterberger, B., 2003. Visitor monitoring methods for managing public use pressures in the Danube Floodplains National Park. *Austria. J. Nat. Conserv.* 11, 260–267. <https://doi.org/10.1078/1617-1381-00057>.
- Belsoy, J., Korir, J., Yego, J., 2012. Environmental impacts of tourism in protected areas. *J. Environ. Earth Sci.* 2, 64–74.
- Buckley, R., 2010. *Conservation Tourism*. CABI, Wallingford.
- Buitrago-Esquinas, E.M., Foronda-Robles, C., Yñiguez-Ovando, R., 2023. A literature review on overtourism to guide the transition to responsible tourism. *Revista de Estudios Andaluces* 45, 45. <https://doi.org/10.12795/rea.2023.i45.04>.
- Butler, R.W., Dodds, R., 2022. Overcoming overtourism: a review of failure. *Tourism Rev.* 77 (1), 35–53. <https://doi.org/10.1108/TR-04-2021-0215>.
- Capocchi, A., Vallone, C., Pierotti, M., Amaduzzi, A., 2019. Overtourism: a literature review to assess implications and future perspectives. *Sustainability* 11, 3303. <https://doi.org/10.3390/su11123303>.
- Cessford, G., Burns, R., 2008. *Monitoring Visitor Numbers in New Zealand National Parks and Protected Areas: A Literature Review and Development Summary*. Science & Technical Pub., Department of Conservation, Wellington.
- Cessford, G., Muhar, A., 2003. Monitoring options for visitor numbers in national parks and natural areas. *J. Nat. Conserv.* 11, 240–250. <https://doi.org/10.1078/1617-1381-00055>.
- Chaney, D., Seraphin, H., 2023. A systematic literature review and lexicometric analysis on overtourism: towards and ambidextrous perspective. *J. Environ. Manag.* 347 (1), 119123. <https://doi.org/10.1016/j.jenvman.2023.119123>.
- Cheng, L., Liu, L., 2022. Exploring posttraumatic growth after the COVID-19 pandemic. *Tourism Manag.* 90, 104474. <https://doi.org/10.1016/j.tourman.2021.104474>.
- Cole, D.N., 2004. Carrying capacity and visitor management: facts, values, and the role of science. In: Harmon, D., Kilgore, B.M., Vietzke, G.E. (Eds.), *Protecting Our Diverse Heritage: the Role of Parks, Protected Areas, and Cultural Sites*, the George Wright Society. *Proceedings of the 2003 George Wright Society/National Park Service Joint Conference*. Michigan, Hancock, pp. 43–46.
- Damjanović, I., 2021. Overtourism and the local community well-being. In: Mandić, A., Petrić, L. (Eds.), *Mediterranean Protected Areas in the Era of Overtourism*.

- Challenges and Solutions. Springer International Publishing, Cham, pp. 93–115. https://doi.org/10.1007/978-3-030-69193-6_5.
- Driml, S., Common, M., 1995. Economic and financial benefits of tourism in major protected areas. *Aust. J. Environ. Manag.* 2, 19–29.
- Eagles, P., McCool, S., Haynes, C., 2002. *Sustainable Tourism in Protected Areas: Guidelines for Planning and Management*. Gland, Cambridge.
- Environment, 2023. (Ochrona Środowiska 2023), 2023. GUS, Warszawa.
- Erlebach, M., Romportl, D., 2021. Časoprostorová distribuce turizmu v Krkonoších a jeho environmentální dopady. *Opera Corcon.* 58, 5–25.
- Farías-Torbidoni, E.I., Dorado, V., Morera, S., Nogueira-Mendes, R., 2023. Optimizing the use of automatic counters to monitor visits to protected natural areas: the case study of Montsant Natural Park, Spain. *Curr. Issues Tourism* 26, 3348–3363. <https://doi.org/10.1080/13683500.2022.2119551>.
- Ferreira, S., Harmse, A., 2014. Kruger National Park: tourism development and issues around the management of large numbers of tourists. *J. Ecotourism* 13, 16–34. <https://doi.org/10.1080/14724049.2014.925907>.
- Fidelus-Orzechowska, J., Sitarz, M., Król, M., 2024. Change patterns between 1993 and 2023 and effects of COVID-19 on tourist traffic in Tatra national park (Poland). *Land* 13, 516. <https://doi.org/10.3390/land13040516>, 2024.
- Ghelichipour, Z., Muhar, A., 2008. Visitor risk management in core zones of protected areas: first results from a survey of European park administrations. In: Raschi, A., Trampetti, S. (Eds.), *Management for Protection and Sustainable Development. Conference Proceedings, Montecatini Terme, Italy*, pp. 14–19. 10.2008.
- Graja-Zwolińska, S., 2009. Rola wskaźnika chłonności turystycznej w kształtowaniu przestrzeni turystycznej parków narodowych. *Studia i Materiały Centrum Edukacji Przyrodniczo-Leśnej* 11 (23), 187–192.
- Gulić, J., Štruc, S., Debeljak Šabec, N., 2014. Monitoring and management of visitors on Pohorje Mountain active raised bogs (Slovenia). In: Reimann, M., Sepp, K., Pärna, K., Tuula, R. (Eds.), *The 7th International Conference on Monitoring and Management of Visitors in Recreational and Protected Areas (MMV) Local Community and Outdoor Recreation*, August 20–23, 2014 Tallinn, Estonia, Conference Proceedings. Hammit, W.E., Cole, D.N., Monz, C.A., 2015. *Wildland Recreation*. Wiley Blackwell.
- Hausser, Y., Travis, T., Finger-Stich, A., 2006. Beyond carrying capacity in recreation management: in search of alternatives. In: Siegrist, D., Clivaz, C., Hunziker, M., Iten, S. (Eds.), *Exploring the Nature of Management. Proceedings of the Third International Conference on Monitoring and Management of Visitor Flows in Recreational and Protected Areas*. University of Applied Sciences Rapperswil, Switzerland, pp. 191–192, 13–17 September 2006, Rapperswil.
- Hernández, M., Baidal, J., Mendoza de Miguel, S., 2019. Overtourism in urban destinations: the myth of smart solutions. *Bol. Asoc. Geógrafos Esp* 83, 1–38. <https://doi.org/10.21138/bage.2830>.
- Hibner, J., Zięba, A., Filar, B., Taczanowska, K., 2023a. Visitors' perception of crowding. A case study from the Tatra Mountains. *Prace Geograficzne* 172, 127–145. <https://doi.org/10.4467/20833113PG.23.018.19205>.
- Hibner, J., Kozumplikova, A., Vasiljević, D., Varga, Á., Morante, T., Švajda, J., Zawilińska, B., Marković, V., Oppong Wiawe, J., Laanisto, L., Casimiro, D., 2023b. VIMOMA : Experience Design and Nature Conservation via Visitor Monitoring and Management in Protected Areas : a Guidebook for Effective Monitoring of Visitors in Protected and Recreational Areas. Institute of Geography and Spatial Management of Jagiellonian University, Kraków.
- Hornback, K.E., Eagles, P.F.J., 1999. *Guidelines for Public Use Measurement and Reporting at Parks and Protected Areas*. IUCN, Gland.
- Horst, L., Taczanowska, K., Porst, F., Arnberger, A., 2023. Evaluation of GNSS-based Volunteered Geographic Information for assessing visitor spatial distribution within protected areas: a case study of the Bavarian Forest National Park, Germany. *Appl. Geogr.* 150, 102825. <https://doi.org/10.1016/j.apgeog.2022.102825>.
- Hu, H., Zhang, J., Wang, C., Yu, P., Chu, G., 2018. What influences tourists' intention to participate in the Zero Litter Initiative in mountainous tourism areas: a case study of Huangshan National Park, China. *Sci. Total Environ.* 657, 1127–1137. <https://doi.org/10.1016/j.scitotenv.2018.12.114>.
- Immoos, U., Hunziker, M., 2015. The effect of communicative and on-site measures on the behaviour of winter sports participants within protected mountain areas – results of a field experiment. *J. Prot. Mt. Areas Res. Manag.* 7, 17–25. <https://doi.org/10.1553/eco.mont-7-1s17>.
- Jancura, P., Beláček, B., Bohálová, I., Slámová, I., 2009. Scientific significance of visualisation methods for expression of selected attributes of landscape character of the High Tatra Mountains. *Landf. Anal.* 10, 44–49.
- Jenkins, J.S., Abatzoglou, J.T., Wilkins, E.J., Perry, E.E., 2023. Visitation to national parks in California shows annual and seasonal change during extreme drought and wet years. *PLOS Clim* 2, e0000260. <https://doi.org/10.1371/journal.pclm.0000260>.
- Job, H., Majewski, L., Engelbauer, M., Bittlingmaier, S., Woltering, M., 2021. Establishing a standard for park visitation analyses: insights from Germany. *J. Outdoor Recreat. Tour.* 35, 100404. <https://doi.org/10.1016/j.jort.2021.100404>.
- Juradó Rota, J., Pérez Albert, M.Y., Serrano Giné, D., 2019. Visitor monitoring in protected areas: an approach to Natura 2000 sites using Volunteered Geographic Information (VGI). *Geogr. Tidsskr.-Dan. J. Geogr.* 119, 69–83. <https://doi.org/10.1080/00167223.2019.1573409>.
- Kajala, L., Almk, A., Dahl, R., Diksaitė, L., Erkkonen, J., Fredman, P., Jensen, F., Karoles, K., Sievänen, T., Skov-Petersen, F., Vistad, O., Wallsten, P., 2007. *Visitor Monitoring in Nature Areas: A Manual Based on Experiences from the Nordic and Baltic Countries*. TemaNord. Swedish Environmental Protection Agency, Stockholm.
- Kohl, I., Pekny, R., 2011. Management in the Dürrenstein Wilderness Area – how much human intervention can the wilderness tolerate? *J. Prot. Mt. Areas Res. Manag.* 1, 51–54. <https://doi.org/10.1553/eco.mont-3-1s51>.
- Krakowiak, B., 2000. A description of the tourist traffic in the west carpathian national parks - the functional types of parks. *Turyzm* 10, 5–37. Available at: <http://hdl.handle.net/11089/28145>. (Accessed 13 August 2024).
- Lebrun, A.-M., Su, C.-J., Bouchet, P., 2021. A more sustainable management of domestic tourists in protected natural parks: a new trend in sport tourism after the covid-19 pandemic? *Sustainability* 13, 7750. <https://doi.org/10.3390/su13147750>.
- Leung, Y.-F., Spenceley, A., Hvenegaard, G., Buckley, R., 2018. *Tourism and Visitor Management in Protected Areas*. IUCN, Gland. <https://doi.org/10.2305/IUCN.CH.2018.PAG.27.en>.
- Levin, N., Lechner, A.M., Brown, G., 2017. An evaluation of crowdsourced information for assessing the visitation and perceived importance of protected areas. *Appl. Geogr.* 79, 115–126. <https://doi.org/10.1016/j.apgeog.2016.12.009>.
- Mandic, A., 2023. Protected area management effectiveness and COVID-19: the case of Plitvice Lakes national park, Croatia. *Journal of Outdoor Recreation and Tourism* 41, 100397. <https://doi.org/10.1016/j.jort.2021.100397>.
- Manning, R.E., Anderson, L.E., 2012. *Managing Outdoor Recreation: Case Studies in the National Parks*. CABI, Cambridge, Mass.
- McCool, S.F., Lime, D.W., 2001. Tourism carrying capacity: tempting fantasy or useful reality? *J. Sustain. Tourism* 9, 372–388. <https://doi.org/10.1080/09669580108667409>.
- McGinlay, J., Gkoumas, V., Holtvoeth, J., Fuertes, R.F.A., Bazhenova, E., Benzoni, A., Botsch, K., Martel, C.C., Sánchez, C.C., Cervera, I., Chaminade, G., Doerstel, J., García, C.J.F., Jones, A., Lammertz, M., Lotman, K., Odar, M., Pastor, T., Ritchie, C., Santi, S., Smolej, M., Rico, F.S., Waterman, H., Zwijacz-Kozica, T., Kontoleon, A., Dimitrakopoulos, P.G., Jones, N., 2020. The impact of COVID-19 on the management of European protected areas and policy implications. *Forests* 11, 1214. <https://doi.org/10.3390/f1111214>.
- Měření návštěvnosti. <https://www.mereninavstevnosti.cz/>, 2024–. (Accessed 28 February 2024).
- Miazek, P., 2019. Turystyka na obszarach chronionych w Polsce. In: *Warsztaty Z Geografii Turyzmu*. Wydawnictwo Uniwersytetu Łódzkiego. <https://doi.org/10.18778/8142-698-5.08>.
- Mika, M., 2024. Społeczności lokalne zależne od turystyki. *Perspektywa ewolucyjna (Tourism-dependent communities. Evolutionary perspectives)*. Instytut Geografii I Gospodarki Przestrzennej. Uniwersytet Jagielloński, Kraków.
- Mitterwallner, V., Peters, A., Edelhoff, H., Mathes, G., Nguyen, H., Peters, W., Heurich, M., Steinbauer, M.J., 2023. Automated visitor and wildlife monitoring with camera traps and machine learning. *Remote Sens. Ecol. Conserv. n/a*. <https://doi.org/10.1002/rse2.367>.
- Monteiro, L., Pereira-da-Silva, C., 2012. Assessing carrying capacity in protected areas trails: the formosinho trail – arrábida natural park (Portugal). In: Fredman, P., Stenseke, M., Liljendahl, L., Mossing, A., Laven, D. (Eds.), *The 6th International Conference on Monitoring and Management of Visitors in Recreational and Protected Areas Outdoor Recreation in Change – Current Knowledge and Future Challenges*. Stockholm, Sweden, August 21–24, 2012. Stockholm.
- Muñoz, L., Hausner, V.H., Monz, C.A., 2019. Advantages and limitations of using mobile apps for protected area monitoring and management. *Soc. Nat. Resour.* 32, 473–488. <https://doi.org/10.1080/08941920.2018.1544680>.
- Oklevik, O., Gössling, S., Hall, C.M., Steen Jacobsen, J.K., Petter, G.I., McCabe, S., 2019. Overtourism, optimisation, and destination performance indicators: a case study of activities in Fjord Norway. *J. Sustain. Tourism* 27, 1804–1824. <https://doi.org/10.1080/09669582.2018.1533020>, 1804–824.
- Papadopoulou, N.M., Ribeiro, M.A., Prayag, G., 2023. Psychological determinants of tourist satisfaction and destination loyalty: the influence of perceived overcrowding and overtourism. *J. Trav. Res.* 62 (3), 644–662. <https://doi.org/10.1177/00472875221089049>.
- Partyka, J., 2010. Ruch turystyczny w polskich parkach narodowych (Tourist traffic in National Park in Poland – in Polish). *Folia Turistica* 22, 9–25. Available at: http://www.folia-turistica.pl/attachments/article/402/FT_22_2010.pdf. (Accessed 13 August 2024).
- Perkumienė, D., Pranskūnienė, R., 2019. Overtourism: between the right to travel and residents' rights. *Sustainability* 11, 2138. <https://doi.org/10.3390/su11072138>.
- Pickering, C.M., Hill, W., Newsome, D., Leung, Y.F., 2010. Comparing hiking, mountain biking and horse riding impacts on vegetation and soils in Australia and the United States of America. *J. Environ. Manag.* 91, 551–562. <https://doi.org/10.1016/j.jenvman.2009.09.025>.
- Plummer, R., Fennell, D.A., 2009. Managing protected areas for sustainable tourism: prospects for adaptive co-management. *J. Sustain. Tourism* 17, 149–168.
- Prato, T., 2001. Modeling carrying capacity for national parks. *Ecol. Econ.* 39 (3), 321–331.
- Prędk, R., 2012. Ruch turystyczny w Bieszczadzkim Parku Narodowym w latach 2009–2011. *Rocz. Bieszczadzkie* 358–377.
- Prędk, R., Demko, T., 2015. Ruch turystyczny w Bieszczadzkim Parku Narodowym w latach 2012–2014. *Rocz. Bieszczadzkie* 23, 367–385.
- Prędk, R., Demko, T., 2018. Ruch turystyczny w Bieszczadzkim Parku Narodowym w latach 2015–2017. *Rocz. Bieszczadzkie* 26, 249–266.
- Prędk, R., Demko, T., 2021. Ruch turystyczny w Bieszczadzkim Parku Narodowym w latach 2018–2020. *Rocz. Bieszczadzkie* 29, 143–158.
- Rączkowska, Z., Kozłowska, A., 2010. Wpływ turystyki na rzeźbę i roślinność przy szczytach w otoczeniu Kasprowego Wierchu. In: Krzan, Z. (Ed.), *Przyroda Tatrzańskiego Parku Narodowego a Człowiek. Nauka a Zarządzenie Obszarem Tatr i Ich Otoczeniem*. Tatrzański Park Narodowy, Zakopane.
- Religa, P., Addach, S., 2020. The problem of solid waste on the tourist trails of Tatra National Park, Poland, *eco. mont – Volume 12*, Number 1, January 2020, ISSN 2073-106X print version – ISSN 2073-1558 online version: <http://epub.oeaw.ac.at/eco.mont> <https://dx.doi.org/10.1553/eco.mont-12-1s35>.

- Rogowski, M., 2019. Assessing the tourism carrying capacity of hiking trails in the Szczeliniec Wielki and Błędne Skały in Stołowe Mts. National Park, Forest Research Papers 80, 125–135. <https://doi.org/10.2478/frp-2019-0011>.
- Rogowski, M., 2020. Monitoring System of tourist traffic (MSTT) for tourists monitoring in mid-mountain national park, SW Poland. *J. Mt. Sci.* 17, 2035–2047. <https://doi.org/10.1007/s11629-019-5965-y>.
- Rogowski, M., 2022a. The impact of Covid-19 pandemic on nature-based tourism in national park. The case of Stołowe Mts. National Park and Karkonosze/Giant Mts. National Park, Poland. *Journal of Environmental Management and Tourism* 13, 572–585. [https://doi.org/10.14505/jemt.v13.2\(58\).25](https://doi.org/10.14505/jemt.v13.2(58).25).
- Rogowski, M., 2022b. Effects of Covid-19 on tourist's behavior and number in mountain national park: the case of the Stołowe Mts. National Park, Poland. *J. Mt. Sci.* 19, 2044–2059. <https://doi.org/10.1007/s11629-021-7205-5>.
- A Method for Overtourism Optimisation for Protected Areas. *Journal of Outdoor Recreation and Tourism*.
- Rogowski, M., Małek, B., 2016. Monitoring ruchu turystycznego w Parku Narodowym Gór Stołowych – założenia projektu i wstępne wyniki badań. In: Młynarczyk, Z., Zajadacz, A. (Eds.), *Uwarunkowania I Planu Rozwoju Turystyki. Turystyka Przyrodnicza I Uwarunkowania Jej Rozwoju*. Bogucki Wydawnictwo Naukowe, Poznań.
- Rogowski, M., Piotrowski, K., 2022. Assessment and accuracy improvement of pyroelectric sensors (Eco-Counter) based on visitors count in national park. The case: monitoring system of tourist traffic in Stołowe mountains national park, Poland. *Environmental and Climate Technologies* 26, 182–198. <https://doi.org/10.2478/rtuect-2022-0015>.
- Santoro, R., Jucker, T., Prisco, I., Carboni, M., Battisti, C., Acosta, A.T.R., 2012. Effects of trampling limitation on coastal dune plant communities. *Environ. Manag.* 49, 534–542. <https://doi.org/10.1007/s00267-012-9809-6>.
- Sæþórsdóttir, A.D., Hall, M., Wendt, M., 2020a. Overtourism in Iceland: fantasy or reality? *Sustainability* 12, 7375. <https://doi.org/10.3390/su12187375>.
- Sæþórsdóttir, A.D., Hall, M., Wendt, M., 2020b. From boiling to frozen? The rise and fall of international tourism to Iceland in the era of overtourism. *Environments* 7, 59. <https://doi.org/10.3390/environments7080059>.
- Schamel, J., Job, H., 2013. Crowding in Germany's national parks: the case of the low mountain range Saxon Switzerland National Park. *Eco Mont - Journal on Protected Mountain Areas Research and Management* 5, 27–34. <https://doi.org/10.1553/eco.mont-5-1s27>.
- Semczuk, M., 2012. Ruch turystyczny w gorczańskim parku narodowym. *Ochr. Beskidów Zachodnich* 4, 98–110.
- Seraphin, H., Sheeran, P., Pilato, M., 2018. Over-tourism and the fall of Venice as a destination. *J. Destin. Market. Manag.* 9, 374–376. <https://doi.org/10.1016/j.jdmm.2018.01.011>.
- Skawinski, P., 2010. Zarządzanie ruchem turystycznym w tatrzańskim parku narodowym (tourism management in the Tatra national park). *Folia Tur* 25–34.
- Smolčić Jurdana, D., Zmijanović, L., 2014. The effect of tourism seasonality on protected areas. In: *Biennial International Congress. Tourism & Hospitality Industry*. University of Rijeka, Faculty of Tourism & Hospitality Management.
- Sobhani, P., Esmailzadeh, H., Sadeghi, S.M.M., Marcu, M.V., 2022. Estimation of ecotourism carrying capacity for sustainable development of protected areas in Iran. *Int. J. Environ. Res. Publ. Health* 19, 1059. <https://doi.org/10.3390/ijerph19031059>.
- Somarrriba-Chang, M., Wallentinus, H.-G., 2012. Recreational carrying capacity in hiking trails. Three case studies in protected areas in Nicaragua. In: Fredman, P., Stenseke, M., Liljendahl, L., Mossing, A., Laven, D. (Eds.), *The 6th International Conference on Monitoring and Management of Visitors in Recreational and Protected Areas Outdoor Recreation in Change – Current Knowledge and Future Challenges*. Stockholm, Sweden, August 21–24, 2012. Stockholm.
- Spychala, A., Graja-Zwolińska, S., 2014. Monitoring ruchu turystycznego w parkach narodowych. *Barom. Reg. Anal. Prognozy* 171–177.
- Staniewska-Zątek, W., 2007. *Turystyka a Przyroda I Jej Ochrona*. Bogucki Wydawnictwo Naukowe, Poznań.
- Stankey, G.H., McCool, S.F., 1984. Carrying capacity in recreational settings: evolution, appraisal and application. *Leisure Sci.* 6 (number 4), 453–473.
- Sterl, P., Wagner, S., Arnberger, A., 2004. Social carrying capacity of canoeists in Austria's Danube Floodplains national park. *Working Papers of the Finnish Forest Research Institute* 2, 256–263.
- Šturcel, M., Švajda, J., 2005. Turystyka w wysokogórskim środowisku Tatr Słowackich. *Návštevnosť vysokohorského prostredia slovenských Tatier*. In: Ładygin, Z., Chovancová, B. (Eds.), *Monitoring Ruchu Turystycznego W Tatrach. Monitoring Návštevnosti Tatier*; TPN, Zakopane, pp. 37–42.
- Švajda, J., Masny, M., Korony, S., Mezei, A., Machar, I., Taczanowska, K., 2018. Visitor profiling using characteristics of socio-demographic and spatial behavior as tools to support the management of protected mountain areas. *Geografie* 123, 461–478. <https://doi.org/10.37040/geografie2018123040461>.
- Tenkanen, H., Di Minin, E., Heikinheimo, V., Hausmann, A., Herbst, M., Kajala, L., Toivonen, T., 2017. Instagram, Flickr, or Twitter: assessing the usability of social media data for visitor monitoring in protected areas. *Sci. Rep.* 7, 17615. <https://doi.org/10.1038/s41598-017-18007-4>.
- Volgger, M., Garcia, C.I., Taplin, R., Pforr, C., 2022. Ability of residents to assess relative risk from tourists during the COVID-19 pandemic. *Tour. Recreat. Res.* 1–6. <https://doi.org/10.1080/02508281.2022.2083844>, 0.
- Walden-Schreiner, C., Rossi, S.D., Barros, A., Pickering, C., Leung, Y.-F., 2018. Using crowd-sourced photos to assess seasonal patterns of visitor use in mountain-protected areas. *Ambio* 47, 781–793. <https://doi.org/10.1007/s13280-018-1020-4>.
- Worboys, G.L., Lockwood, M., Kothari, A., Feary, S., Pulsford, I. (Eds.), 2015. *Protected Area Governance and Management*, first ed. ANU Press. <https://doi.org/10.22459/PAGM.04.2015>.
- Zaręba, D., 2010. *Ekoturystyka (Ecotourism)*. PWN, Warszawa.
- Zawilińska, B., 2021. Metody badania ruchu turystycznego i konsumpcji usług turystycznych w polskich parkach narodowych. *Stud. Ind. Geogr. Comm. Pol. Geogr. Soc.* 35, 41–61.
- Zawilińska, B., 2023. Apartament z widokiem na apartament z widokiem. Współczesny rozwój bazy noclegowej w powiecie tatrzańskim z perspektywy lokalnej społeczności. *Pr. Geogr.* 2023, 125–147. <https://doi.org/10.4467/20833113PG.23.012.18112>.
- Zawilińska, B., Mika, M., 2023. Wydatki turystów odwiedzających Tatrzański Park Narodowy: wielkość, struktura oraz efekty ekonomiczne w lokalnej gospodarce. In: *Park Narodowy: Funkcja Ekonomiczna a Postawy Społeczności Lokalnej*. Instytut Geografii I Gospodarki Przestrzennej. Uniwersytet Jagielloński, Kraków.
- Żemła, M., 2024. European historic cities and overtourism – conflicts and development paths in the light of systematic literature review. *International Journal of Tourism Cities* 10 (1), 353–372. <https://doi.org/10.1108/IJTC-07-2023-0140>.
- Zhong, L., Deng, J., Ding, Z., Song P., 2011. Research on environmental impacts of tourism in China: progress and prospect. *J. Environ. Manag.* 92, 2972–2983. <https://doi.org/10.1016/j.jenvman.2011.07.011>.
- Þórhallsdóttir, R., Ólafsson, 2017. A method to analyse seasonality in the distribution of tourists in Iceland. *Journal of Outdoor Recreation and Tourism* 19, 17–24. <https://doi.org/10.1016/j.jort.2017.05.001>.